

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT Plan the learning first; let the checklist follow. Click a field to type • click a box to tick it.

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 6 (11-15 Oct 2026) - Period 2	Unit 2 - 2.2 Advance Photo Editing (1/2)
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 6 / Lesson 2 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Postcards: rubric average level 3/4; export settings weak (wrong file type). Today: explicit export routine and file-type comparison.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students improve colour (Colors > Auto > Color Enhance; Colors > Color Balance) and straighten a tilted image (Measure tool + Straighten), then export as .png. Sprint 2.2 obj. 1-2; Challenge 1; CSTA 3A-DA-09.

Key knowledge / skill / understanding:

Colour = red, green, blue values; Color Balance shifts cyan-red, magenta-green, yellow-blue; straightening rotates by a measured angle.

Success criteria — what will successful learning look / sound like?

I can adjust shadows/midtones/highlights with Color Balance and explain the slider pairs; I can straighten a tilted horizon; I export a .png.

Likely misconception / access or challenge need:

Brightness = colour balance; rotating by eye rather than measuring the angle.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Inquiry / investigation ☐ Time: 10 min	Strategy: Modelling / worked exam ☐ Time: 18 min	Strategy: Think-pair-share ☐ Time: 8 min
What students will do: Investigate: move each Color Balance slider on a test photo for Shadows, Midtones, Highlights; record in a table what happens to the image and to RGB values (Pointer dialog).	What students will do: Follow the worked example then Challenge 1: apply Color Enhance and Color Balance to a dull photo; then use the Measure tool to straighten a tilted horizon photo. Export both as .png with a clear file name.	What students will do: Show 3 versions of the same photo exported as .png, .jpg and .gif. Predict the file sizes, then check real sizes and discuss which format suits which use.
Evidence tool: Portfolio / notebook work ☐ Assessment: ☒ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group	Evidence tool: Practical / performance task ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Mini-whiteboards ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group
Resources: Test image; table template in notebook; Windows > Dockable Dialogs > Pointer.	Resources: GIMP; dull & tilted images from pack; QR videos aCT4, hCYG.	Resources: 3 exported files with sizes; mini-whiteboards.
Support (low performers / SEN): Partially completed table with one example row.	Support (low performers / SEN): Printed screenshots for Measure tool steps; images already open.	Support (low performers / SEN): Size values written as KB with a number line.
Stretch (fast learners / gifted): Explain why yellow-blue slider changes the blue channel value.	Stretch (fast learners / gifted): Straighten using two different methods and compare quality (crop vs. clip).	Stretch (fast learners / gifted): Calculate the % difference between the largest and smallest file.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

- ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared
- ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources
- ☒ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: a dull, tilted holiday photo - 'What is wrong and how can we fix it?'. Model Color Enhance (one click) vs Color Balance (controlled). Think aloud about the three slider pairs. Model the Measure tool: drag along the horizon, click Straighten. Model the export routine: File > Export As > choose type > Export.

What is new or innovative in this lesson? (ETEC looks for this)

Students collect evidence (RGB readings, file sizes) like scientists; numeracy built in through % file-size comparison.

Progress recorded / shared:

- ☒ Gradebook ☐ Parents ☒ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

- ☒ Re-teach ☒ Regroup ☒ Give extra support
- ☒ Stretch ☒ Have students redo / improve

What I will watch for / likely responses:

Using Brightness-Contrast instead of Color Balance; forgetting to choose a range (shadows/midtones/highlights); measuring a vertical line; 'Save' vs 'Export' confusion. Likely: students over-saturate colour - ask 'Does it still look natural?'

Close / exit check (summative + time for learners' questions) Time: 8 min

Exit ticket (3 Qs on Forms): which menu holds Color Enhance; what the three ranges are; which file type was largest and why. Time for learners' questions.

Homework / platform (e.g. Madrasati)

Madrasati: upload before/after colour-corrected photo (.png) and a 2-line explanation of the slider pairs used. Watch QR video 'Remove red eye' (WCUC).

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Investigation table made students notice the RGB link; most could explain one slider pair. Measure tool worked well once modelled twice.

What will I change next time?

Start the file-size task with a visual bar chart - several students struggled converting KB/MB. Check export routine on the board as a poster.

Students needing follow-up:

Students in 10/2 and 10/6 who missed Week 5 need a catch-up sheet for selection tools; 2 EAL students need the slider vocabulary card.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☒ 1 ☐ 2 ☐ 3 ☒ C ☐ H	☒ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☒ 2 ☐ 3 ☐ C ☒ H	☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H
Create something new · 2-1-1-7	Discussion expressing ideas · 2-1-1-8	Pair / group cooperative work · 2-1-1-8	Learner choice independent work · 3-2-1-6	Self-assessment 2-2-1-1	Peer assessment 2-2-1-1	SEN / low specific support · 2-1-1-3	Gifted specific stretch · 2-1-1-3
☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback